Lab Assignment 2: Practicing DDL and DML command
Introduction to Databases (CSE 3151)
Programme: B.Tech. (CSE & CSIT) Semester: 6th

Objective: To understand and apply DDL and DML commands for modifying database structure and manipulating data in a Banking database.

1. Write the SQL expressions for the following SQL queries and verify it in your PC by executing it.

a) Display the structure of all the tables created in Assignment (1).

```
mysql> DESC BRANCH;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| BRANCH_CODE | varchar(5)  | YES  |     | NULL    |       |
| BRANCH_NAME | varchar(30) | YES  |     | NULL    |       |
| CITY        | varchar(20) | YES  |     | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

```
mysql> DESC CUSTOMER;
+-----------+---------------+------+-----+---------+-------+
| Field     | Type          | Null | Key | Default | Extra |
+-----------+---------------+------+-----+---------+-------+
| CUST_NO   | varchar(5)    | YES  |     | NULL    |       |
| CUST_NAME | varchar(30)   | YES  |     | NULL    |       |
| ADDRESS   | varchar(40)   | YES  |     | NULL    |       |
| PHONE     | decimal(10,0) | YES  |     | NULL    |       |
+-----------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)

mysql> DESC ACCOUNT;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| ACC_NO      | varchar(10)   | YES  |     | NULL    |       |
| CUST_NO     | varchar(5)    | YES  |     | NULL    |       |
| BRANCH_CODE | varchar(5)    | YES  |     | NULL    |       |
| ACC_TYPE    | varchar(10)   | YES  |     | NULL    |       |
| BALANCE     | decimal(10,0) | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

```
mysql> DESC LOAN;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| LOAN_NO     | varchar(10)   | YES  |     | NULL    |       |
| CUST_NO     | varchar(5)    | YES  |     | NULL    |       |
| BRANCH_CODE | varchar(5)    | YES  |     | NULL    |       |
| AMOUNT      | decimal(10,0) | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

```
mysql> DESC TRANSACTION_DETAILS;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| TRANS_ID      | varchar(10)   | YES  |     | NULL    |       |
| ACC_NO        | varchar(10)   | YES  |     | NULL    |       |
| TRANS_TYPE    | varchar(10)   | YES  |     | NULL    |       |
| TRANS_AMOUNT  | decimal(10,0) | YES  |     | NULL    |       |
| TRANS_DATE    | date          | YES  |     | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)
```

b) Display the contents of all the tables created in Assignment (1).

```
mysql> SELECT * FROM BRANCH ;
+-------------+----------------+-----------+
| BRANCH_CODE | BRANCH_NAME    | CITY      |
+-------------+----------------+-----------+
| B001        | Main Branch    | Delhi     |
| B002        | City Branch    | Mumbai    |
| B003        | Town Branch    | Chennai   |
| B004        | Central Branch | Delhi     |
| B005        | Market Branch  | Bengaluru |
+-------------+----------------+-----------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM CUSTOMER;
+---------+--------------+-----------+------------+
| CUST_NO | CUST_NAME    | ADDRESS   | PHONE      |
+---------+--------------+-----------+------------+
| C001    | Ankit Kumar  | Delhi     | 9876543210 |
| C002    | Anjali Verma | Mumbai    | 9123456780 |
| C003    | Suresh Patel | Ahmedabad | 9988776655 |
| C004    | Neha Sharma  | Jaipur    | 9090909090 |
| C005    | Amit Singh   | Lucknow   | 8899776655 |
+---------+--------------+-----------+------------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM ACCOUNT;
+--------+---------+-------------+----------+---------+
| ACC_NO | CUST_NO | BRANCH_CODE | ACC_TYPE | BALANCE |
+--------+---------+-------------+----------+---------+
| A1001  | C001    | B001        | Savings  |   25000 |
| A1002  | C002    | B002        | Current  |   40000 |
| A1003  | C003    | B003        | Savings  |   18000 |
| A1004  | C004    | B004        | Savings  |   32000 |
| A1005  | C005    | B005        | Current  |   50000 |
+--------+---------+-------------+----------+---------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM LOAN;
+---------+---------+-------------+--------+
| LOAN_NO | CUST_NO | BRANCH_CODE | AMOUNT |
+---------+---------+-------------+--------+
| L2001   | C001    | B001        | 150000 |
| L2002   | C002    | B002        | 200000 |
| L2003   | C003    | B003        | 120000 |
| L2004   | C004    | B004        | 180000 |
| L2005   | C005    | B005        | 250000 |
+---------+---------+-------------+--------+
5 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM TRANSACTION_DETAILS;
+----------+--------+------------+--------------+------------+
| TRANS_ID | ACC_NO | TRANS_TYPE | TRANS_AMOUNT | TRANS_DATE |
+----------+--------+------------+--------------+------------+
| T001     | A1001  | Deposit    |         5000 | 2025-01-10 |
| T002     | A1002  | Withdraw   |         3000 | 2025-01-12 |
| T003     | A1003  | Deposit    |         7000 | 2025-10-15 |
| T004     | A1004  | Withdraw   |         2000 | 2024-01-18 |
| T005     | A1005  | Deposit    |        10000 | 2024-03-20 |
+----------+--------+------------+--------------+------------+
5 rows in set (0.00 sec)
```

c) Display the customer number and branch code of each account holder.

```
mysql> SELECT CUST_NO , BRANCH_CODE
    -> FROM ACCOUNT ;
+---------+-------------+
| CUST_NO | BRANCH_CODE |
+---------+-------------+
| C001    | B001        |
| C002    | B002        |
| C003    | B003        |
| C004    | B004        |
| C005    | B005        |
+---------+-------------+
5 rows in set (0.01 sec)
```

d) Display the customer number and account balance of customers having Savings accounts.

```
mysql> SELECT CUST_NO , BALANCE
    -> FROM ACCOUNT
    -> WHERE ACC_TYPE = 'Savings';
+---------+---------+
| CUST_NO | BALANCE |
+---------+---------+
| C001    |   25000 |
| C003    |   18000 |
| C004    |   32000 |
+---------+---------+
3 rows in set (0.00 sec)
```

e) Display the account number and transaction ID of customers who have made deposit transactions with a transaction amount greater than 5000.

```
mysql> SELECT ACC_NO , TRANS_ID
    -> FROM TRANSACTION_DETAILS
    -> WHERE TRANS_TYPE = 'Deposit'
    -> AND TRANS_AMOUNT > 5000;
+--------+----------+
| ACC_NO | TRANS_ID |
+--------+----------+
| A1003  | T003     |
| A1005  | T005     |
+--------+----------+
2 rows in set (0.00 sec)
```

f) Display the records of customers whose loan amount is less than 200000.

```
mysql> SELECT *
    -> FROM LOAN
    -> WHERE AMOUNT < 200000;
+---------+---------+-------------+--------+
| LOAN_NO | CUST_NO | BRANCH_CODE | AMOUNT |
+---------+---------+-------------+--------+
| L2001   | C001    | B001        | 150000 |
| L2003   | C003    | B003        | 120000 |
| L2004   | C004    | B004        | 180000 |
+---------+---------+-------------+--------+
3 rows in set (0.00 sec)
```

g) Display the customer numbers of customers who do not have Savings accounts.

```
mysql> SELECT CUST_NO
    -> FROM ACCOUNT
    -> WHERE ACC_TYPE != 'Savings';
+---------+
| CUST_NO |
+---------+
| C002    |
| C005    |
+---------+
2 rows in set (0.00 sec)
```

h) Display the names of different cities distinctly available in the branch table.

```
mysql> SELECT DISTINCT CITY
    -> FROM BRANCH;
+-----------+
| CITY      |
+-----------+
| Delhi     |
| Mumbai    |
| Chennai   |
| Bengaluru |
+-----------+
4 rows in set (0.00 sec)
```

i) Display all account numbers that have transactions in the month of January 2025.

```
mysql> SELECT ACC_NO
    -> FROM TRANSACTION_DETAILS
    -> WHERE TRANS_DATE BETWEEN  '2025-01-01' AND '2025-01-31';
+--------+
| ACC_NO |
+--------+
| A1001  |
| A1002  |
+--------+
2 rows in set (0.00 sec)
```

j) Display account numbers whose balance is not equal to 32000.

```
mysql> SELECT ACC_NO
    -> FROM ACCOUNT
    -> WHERE BALANCE != 32000;
+--------+
| ACC_NO |
+--------+
| A1001  |
| A1002  |
| A1003  |
| A1005  |
+--------+
4 rows in set (0.00 sec)
```

k) Display all columns of the account table sorted in descending order of account balance.

```
mysql> SELECT *
    -> FROM ACCOUNT
    -> ORDER BY BALANCE DESC;
+--------+---------+-------------+----------+---------+
| ACC_NO | CUST_NO | BRANCH_CODE | ACC_TYPE | BALANCE |
+--------+---------+-------------+----------+---------+
| A1005  | C005    | B005        | Current  |   50000 |
| A1002  | C002    | B002        | Current  |   40000 |
| A1004  | C004    | B004        | Savings  |   32000 |
| A1001  | C001    | B001        | Savings  |   25000 |
| A1003  | C003    | B003        | Savings  |   18000 |
+--------+---------+-------------+----------+---------+
5 rows in set (0.00 sec)
```

l) Add a new column EMAIL to the customer table.

```
mysql> ALTER TABLE CUSTOMER
    -> ADD EMAIL VARCHAR(40);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

m) Insert email values into the existing rows of the customer table.

```
mysql> UPDATE CUSTOMER
    -> SET EMAIL = 'ankit@gmail.com'
    -> WHERE CUST_NO  = 'C001';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE CUSTOMER
    -> SET EMAIL='anjali@gmail.com'
    -> WHERE CUST_NO='C002';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE CUSTOMER
    -> SET EMAIL='suresh@gmail.com'
    -> WHERE CUST_NO='C003';
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE CUSTOMER
    -> SET EMAIL='suresh@gmail.com'
    -> WHERE CUST_NO='C003';
Query OK, 0 rows affected (0.00 sec)
Rows matched: 1  Changed: 0  Warnings: 0

mysql>
mysql> UPDATE CUSTOMER
    -> SET EMAIL='amit@gmail.com'
    -> WHERE CUST_NO='C005';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

n) Change the column name CITY to LOCATION in the branch table.

```
mysql> ALTER TABLE BRANCH
    -> RENAME COLUMN CITY TO LOCATION;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

o) Change the name of the table TRANSACTION_DETAILS to BANK_TRANSACTION.

```
mysql> ALTER TABLE TRANSACTION_DETAILS
    -> RENAME TO BANK_TRANSACTION;
Query OK, 0 rows affected (0.02 sec)
```

p) Change the column name ACC_NO to ACCOUNT_NO in the account table.

```
mysql> ALTER TABLE ACCOUNT
    -> RENAME COLUMN ACC_NO TO ACCOUNT_NO;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

q) Change the data type of CUST_NAME column to VARCHAR (50) in the customer table.

```
mysql> ALTER TABLE CUSTOMER
    -> MODIFY CUST_NAME VARCHAR(50);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

r) Change the name of the ACCOUNT table to CUSTOMER_ACCOUNT.

```
mysql> ALTER TABLE ACCOUNT
    -> RENAME TO CUSTOMER_ACCOUNT;
Query OK, 0 rows affected (0.02 sec)
```

s) Change the column size of BRANCH_NAME in the branch table from 30 to 40.

```
mysql> ALTER TABLE BRANCH
    -> MODIFY BRANCH_NAME VARCHAR(40);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

t) Delete all the contents of the BANK_TRANSACTION table.

```
mysql> DELETE FROM BANK_TRANSACTION;
Query OK, 5 rows affected (0.01 sec)
```